Paper 2

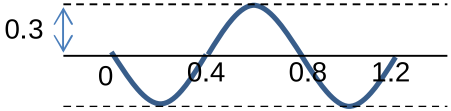
1 **(a)** Hooke's law is obeyed because extension of spring is proportional to force applied. This is shown by the straight line graph of Fig 1.1.

(b) Work done = area under the F-extension graph

$$\text{Work done} = \frac{1}{2} \times (30-12)(1.4) \times 10^{-2} = 0.13\ \text{J}$$

(c) **(i)** $$v_{\text{max}} = \omega x_o = \frac{2\pi}{T} \times (34-30) \times 10^{-2} = \frac{2\pi}{0.84} \times 4 \times 10^{-2} = 0.30\ \text{m s}^{-1}$$

(ii)



(d) **(i)** At equilibrium,

$$W_M = T + U$$

$$T = W - \rho V g = 1.4 - (1000)(\frac{20}{10^6})(9.81) = 1.2\ \text{N}$$

From Fig 1.1, length of spring = 0.275 m

(ii) 1. The oscillation now experiences greater damping hence amplitude will decrease with time.

2. Equilibrium position is at a new position when length of spring is 0.275 m